

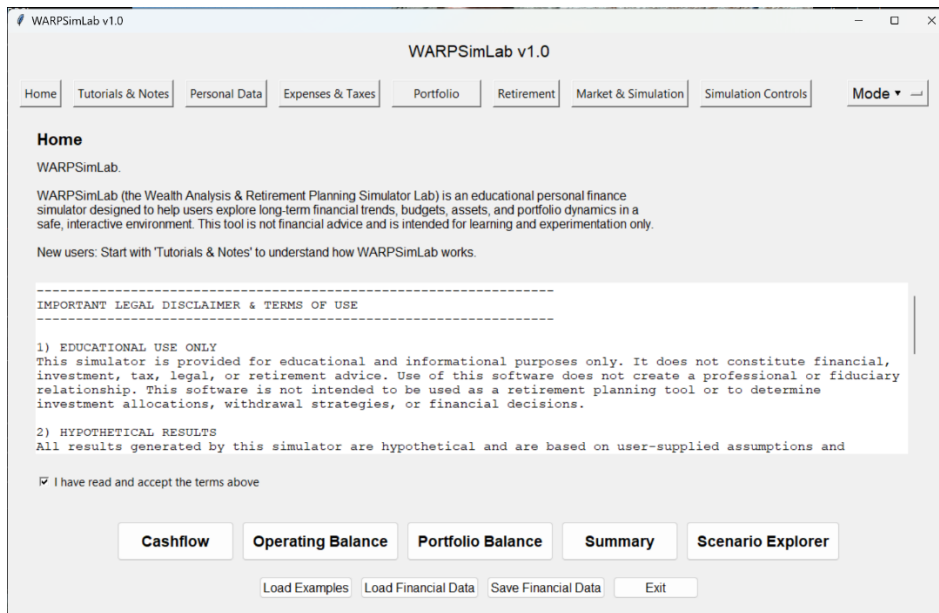
WARPSimLab v1.0

Getting Started – Advanced Mode

1. Overview

Advanced Mode unlocks the full simulation environment. All modeling tabs are enabled, and you gain control over portfolio structure, retirement withdrawal behavior, tax configuration, and market simulation settings.

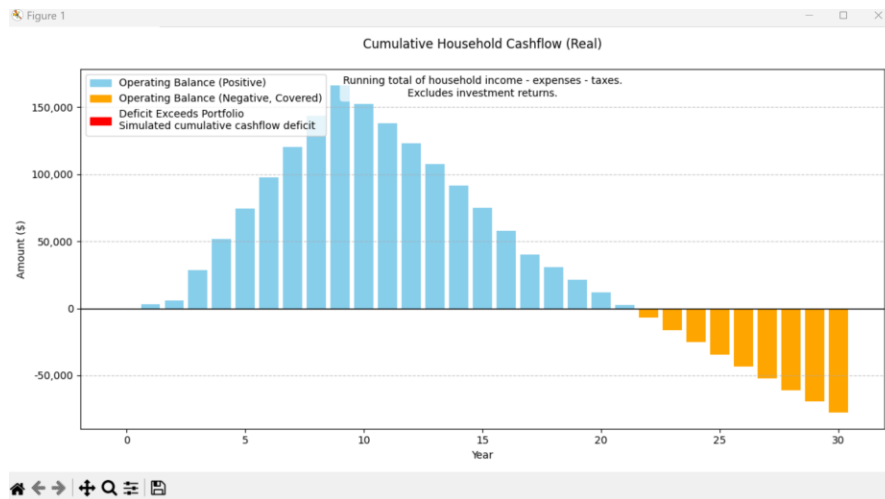
Basic Mode focuses on core cause-and-effect learning. Advanced Mode adds realism, flexibility, and simulation depth.



2. New Visualization Tools

Operating Balance

Operating Balance displays the cumulative net surplus or deficit over time. While the Cashflow chart shows annual income and net results, Operating Balance shows the running total of those yearly outcomes.

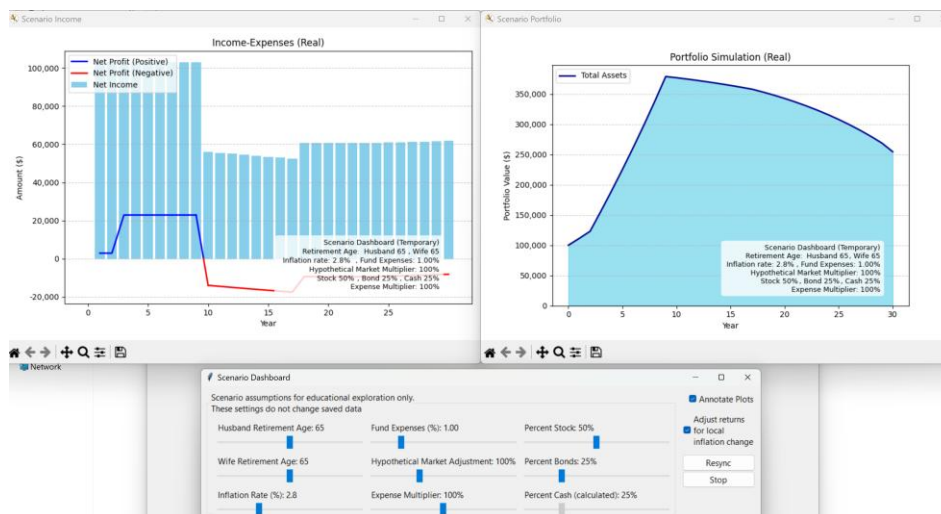


Scenario Explorer

Scenario Explorer opens a separate dashboard with interactive sliders that temporarily adjust key inputs such as:

- Retirement ages
- Fund expense impacts
- Market adjustment factors
- Asset allocation
- Withdrawal or expense multipliers

Changes update plots in real time without modifying saved baseline data. This tool is designed for rapid “what-if” exploration.



3. The Tabs in Advanced Mode

All configuration tabs are enabled in Advanced Mode.

Personal Data (Expanded)

Advanced Mode adds fields for 401(k) contributions/matching, pension income, and annuity income. If your pension includes cost-of-living adjustments, use the pension inflation adjustment setting to model partial inflation indexing. Annuities are typically constant payments; by default the simulator treats annuity income as not inflation adjusted.

Expenses & Taxes (Expanded)

The Expenses panel functions as in Basic Mode. Advanced Mode enables tax controls. Note that tax calculations are approximate. Social Security may be treated as fully taxable in this simplified model.

Portfolio

In Advanced Mode, Starting Assets becomes the full Portfolio editor, where you can enter stocks, bonds, and cash across pre-tax and post-tax accounts (and optionally real estate). The optional Derived Statistics panel summarizes totals and percentage breakdowns to help you review allocation and rebalance targets.

Retirement

This tab controls how retirement is modeled, including whether the simulation treats spending as explicit expenses or uses a withdrawal strategy (for example, a percentage-based withdrawal that can be inflation-adjusted). It also controls options such as including Required Minimum Distributions (RMDs).

Market & Simulation

This tab combines historical market assumptions (expected returns, volatility, inflation) with simulation settings that control how the portfolio is modeled. You can select preset historical datasets or enter assumptions manually. Simulation settings also include the number of simulations, fund expense modeling, and rebalancing options.

Simulation Controls

Simulation Controls manage how results are displayed and exported. These controls affect presentation and comparison, not underlying inputs. See section 4 for more detail.

4. Running Advanced Simulations

Advanced Mode supports multiple simulation styles. These are controlled in the Simulation Controls Tab.

Real/ Nominal Mode

This option controls how dollar values are displayed. **Real** dollars are inflation-adjusted to the start-year purchasing power. **Nominal** dollars are shown in future (inflated) dollars for each year.

Subplot Mode

Subplot Mode determines how results are generated and displayed. Options include:

- Monte Carlo simulations (multiple randomized runs with percentile results)
- Category views, such as income or portfolio detail breakdowns

When Monte Carlo is selected, multiple simulations are run and percentile bands are displayed.

Overlays

Overlays add comparison or reference information to plots—for example expenses, a retirement marker, and ‘no-tax’ or ‘no-fund-expense’ comparisons (when enabled).

5. Experiment and Compare

Advanced Mode is designed for structured experimentation and rewards incremental adjustments.

Adjust one setting at a time—such as asset allocation, withdrawal strategy, inflation assumptions, or simulation style—and rerun the reports. Use Subplot Mode, Monte Carlo, and Overlays to evaluate how changes affect sustainability and variability.

Scenario Explorer provides a faster way to test temporary adjustments without modifying your saved baseline configuration.